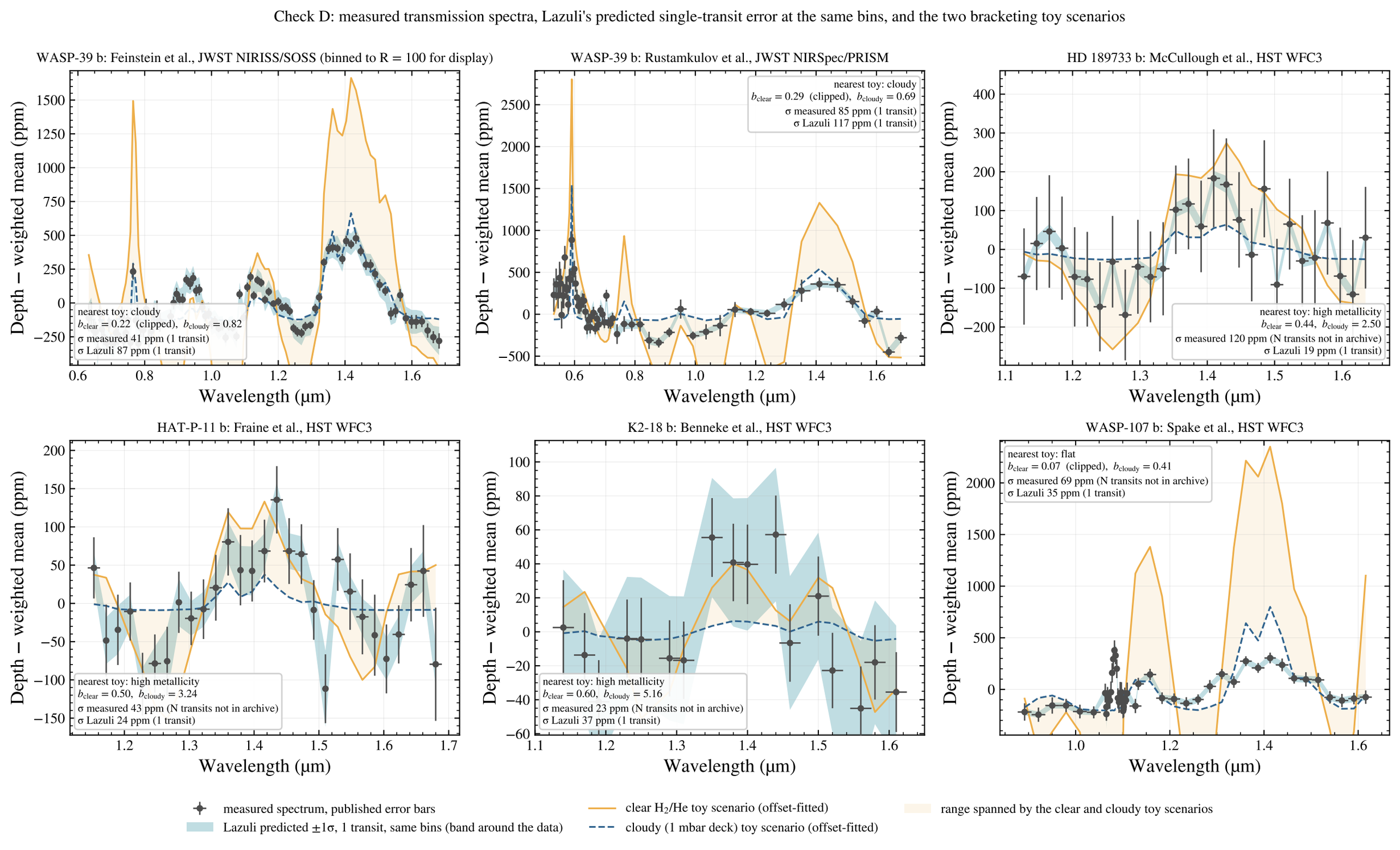

Check D: measured transmission spectra, Lazuli's predicted single-transit error at the same bins, and the two bracketing toy scenarios
WASP-39 b: Feinstein et al., JWST NIRISS/SOSS (binned to R = 100 for display)
nearest toy: cloudy
b_{clear} = 0.22 (clipped), b_{cloudy} = 0.82
σ measured 41 ppm (1 transit)
σ Lazuli 87 ppm (1 transit)
WASP-39 b: Rustamkulov et al., JWST NIRSpec/PRISM
nearest toy: cloudy
b_{clear} = 0.29 (clipped), b_{cloudy} = 0.69
σ measured 85 ppm (1 transit)
σ Lazuli 117 ppm (1 transit)
HD 189733 b: McCullough et al., HST WFC3
nearest toy: high metallicity
b_{clear} = 0.44, b_{cloudy} = 2.50
σ measured 120 ppm (N transits not in archive)
σ Lazuli 19 ppm (1 transit)
HAT-P-11 b: Fraine et al., HST WFC3
nearest toy: high metallicity
b_{clear} = 0.50, b_{cloudy} = 3.24
σ measured 43 ppm (N transits not in archive)
σ Lazuli 24 ppm (1 transit)
K2-18 b: Benneke et al., HST WFC3
nearest toy: high metallicity
b_{clear} = 0.60, b_{cloudy} = 5.16
σ measured 23 ppm (N transits not in archive)
σ Lazuli 37 ppm (1 transit)
WASP-107 b: Spake et al., HST WFC3
nearest toy: flat
b_{clear} = 0.07 (clipped), b_{cloudy} = 0.41
σ measured 69 ppm (N transits not in archive)
σ Lazuli 35 ppm (1 transit)
Depth − weighted mean (ppm)
Wavelength (μm)
measured spectrum, published error bars
Lazuli predicted ±1σ, 1 transit, same bins (band around the data)
clear H_2/He toy scenario (offset-fitted)
cloudy (1 mbar deck) toy scenario (offset-fitted)
range spanned by the clear and cloudy toy scenarios